6–Regression-Models

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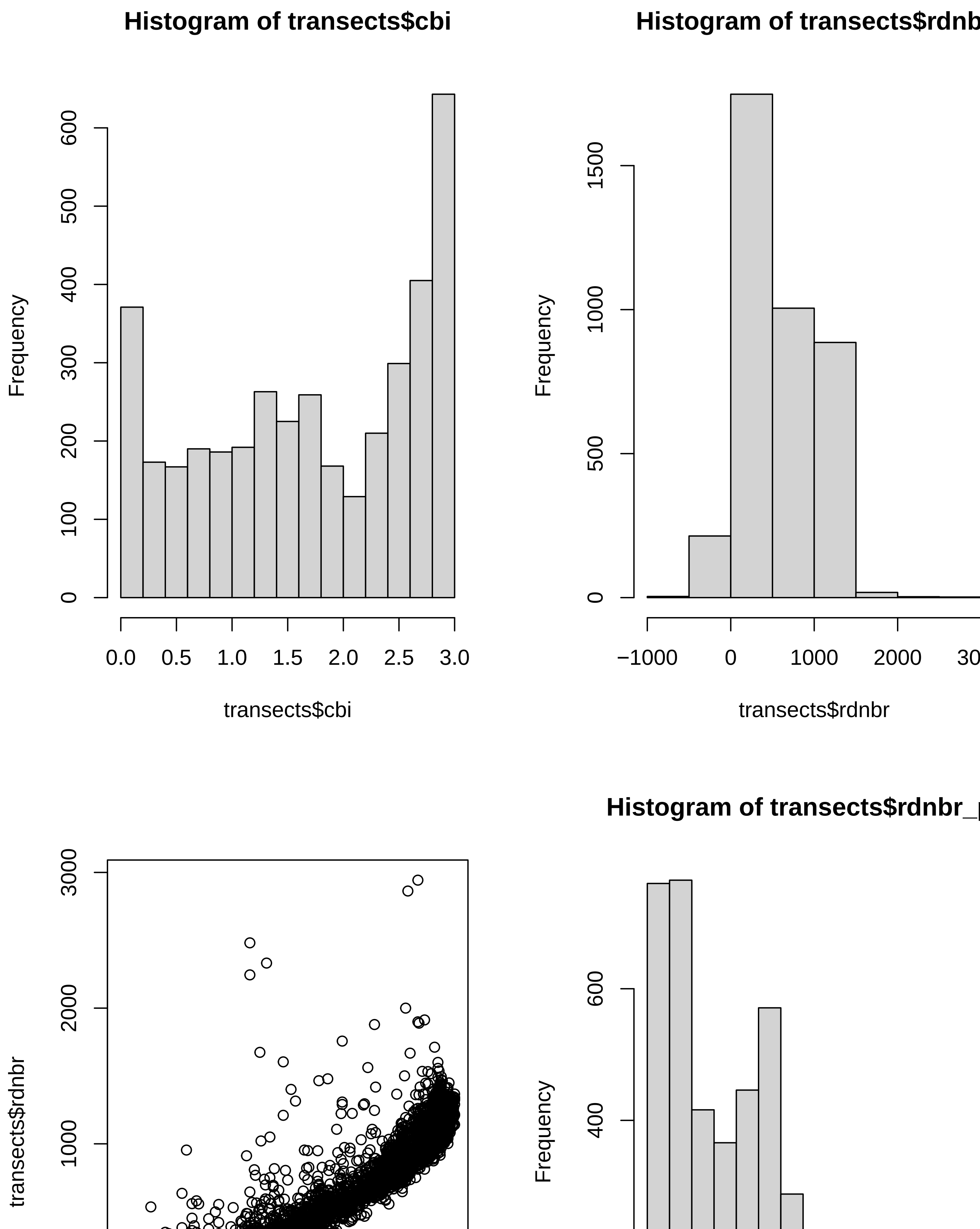
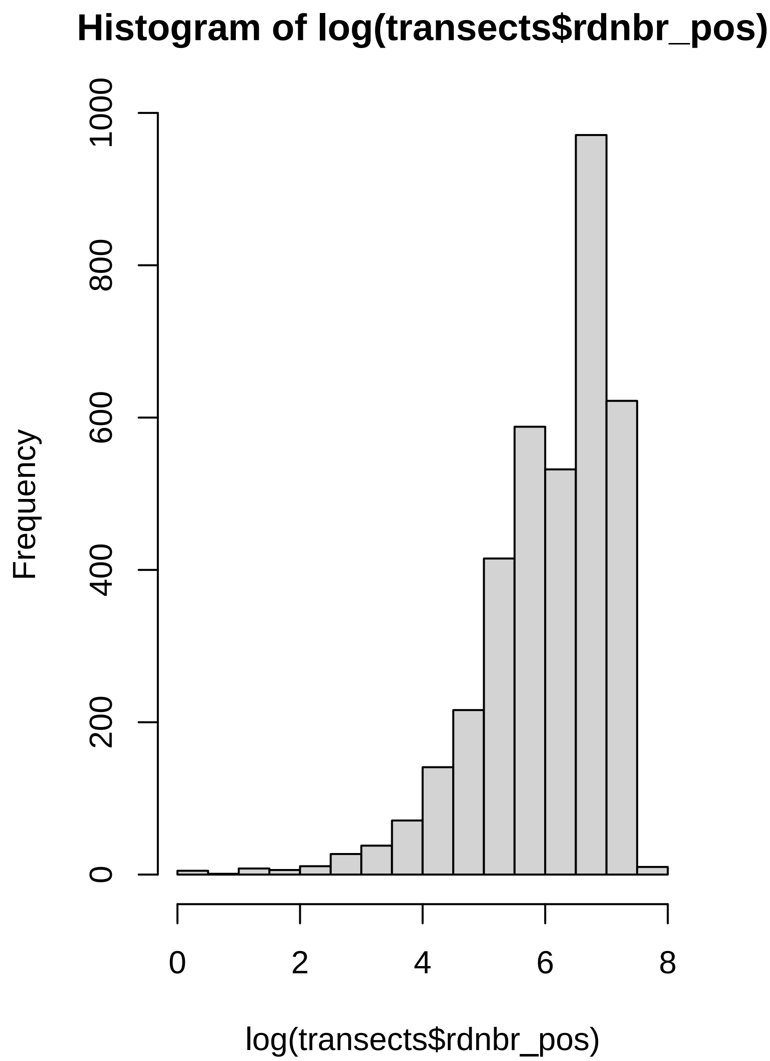
Setup

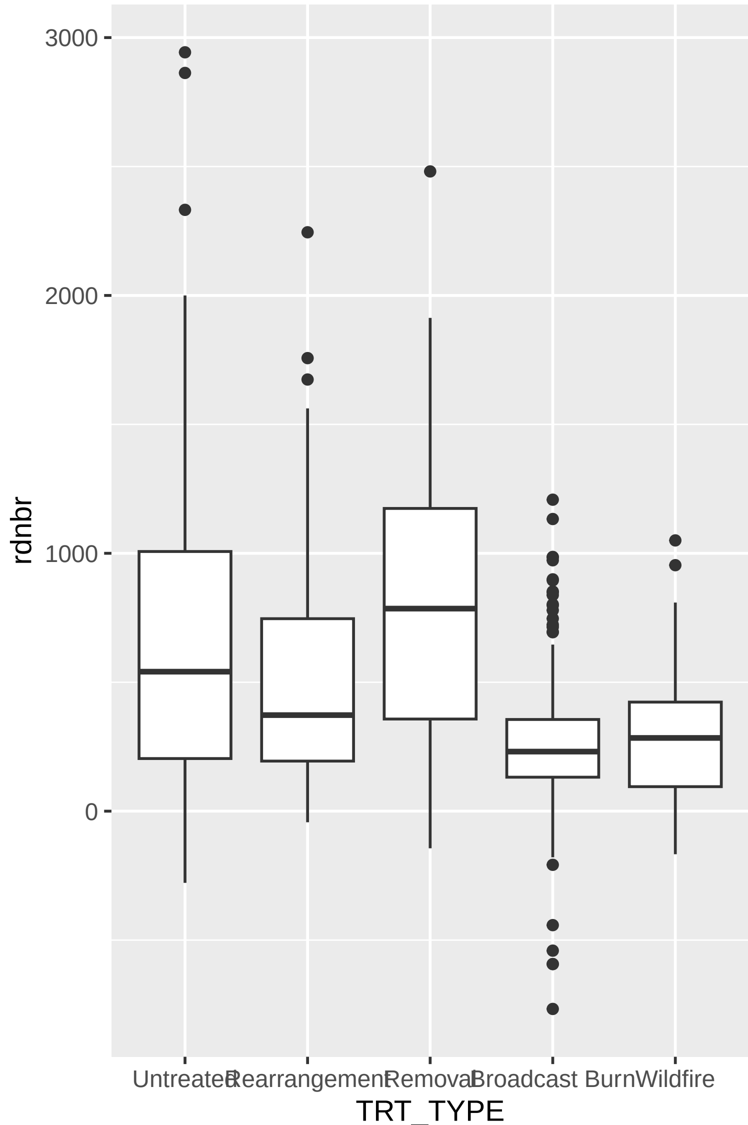
```
## Reading layer ‘transectsDS’ from data source
##   ‘C:\Users\Elizabeth Buhr\OneDrive\Documents\fuels-analysis\data\processed\fuels_analysis.gpkg’
##   using driver ‘GPKG’
## Simple feature collection with 3934 features and 27 fields
## Geometry type: POINT
## Dimension:     XY
## Bounding box:  xmin: 173613.9 ymin: 4094767 xmax: 555237.8 ymax: 4512971
## Projected CRS: NAD83 / UTM zone 13N
```

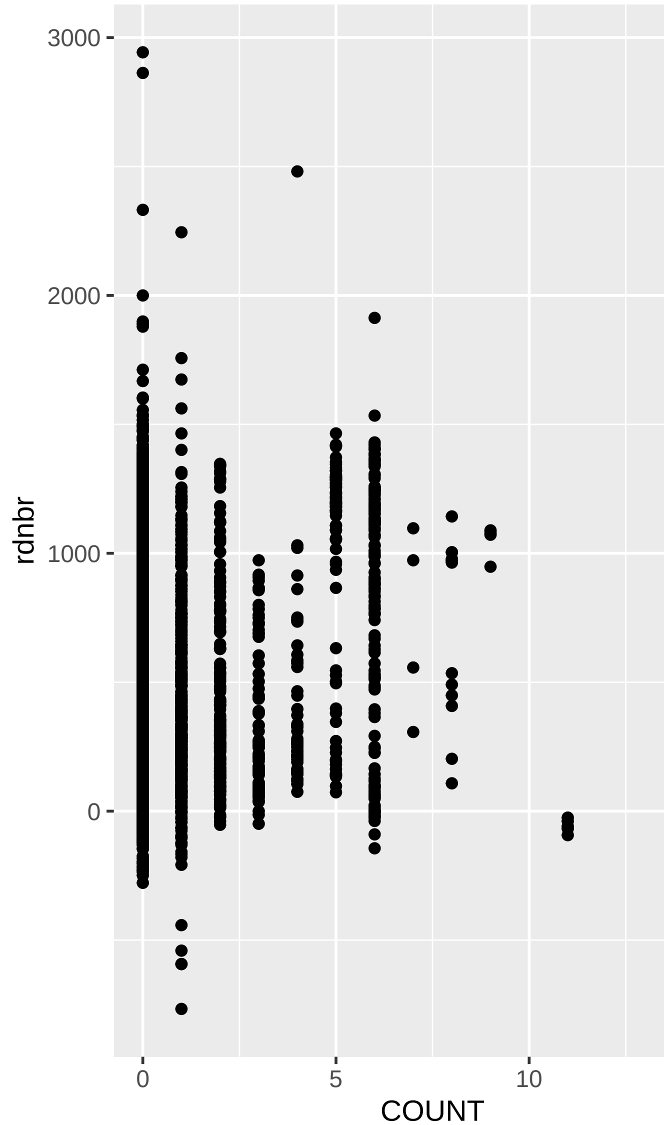
Response Variable Distribution

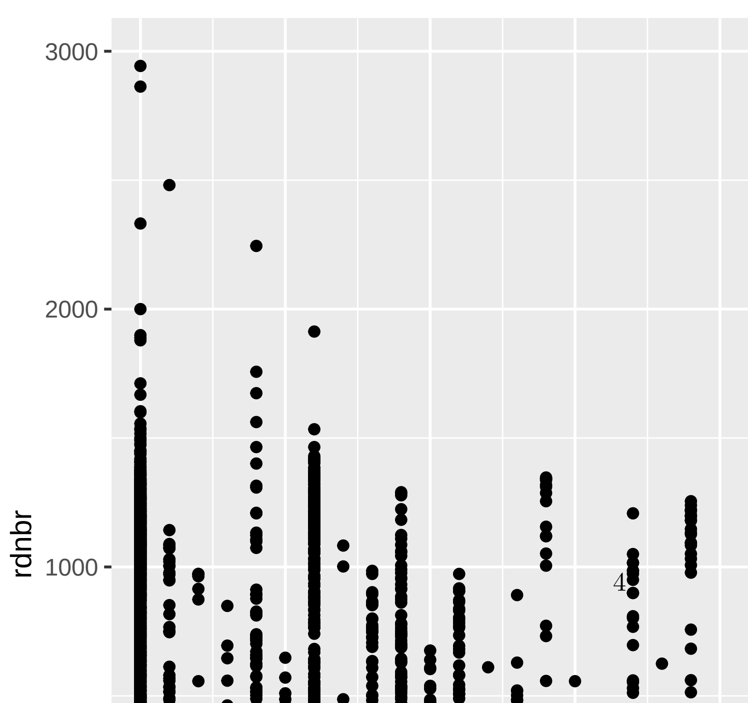


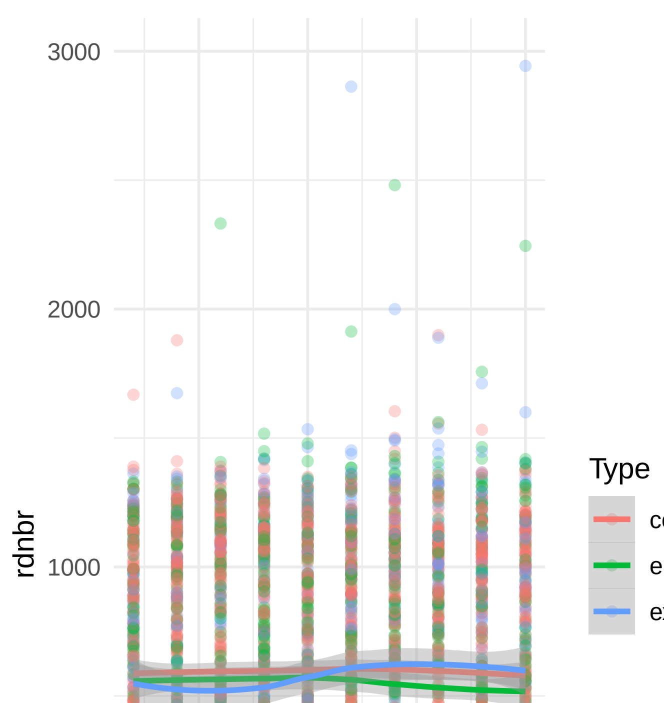

Histogram of log(transects$rdnbr_pos)
Frequency
0
200
400
600
800
1000
0
2
4
6
8
log(transects$rdnbr_pos)

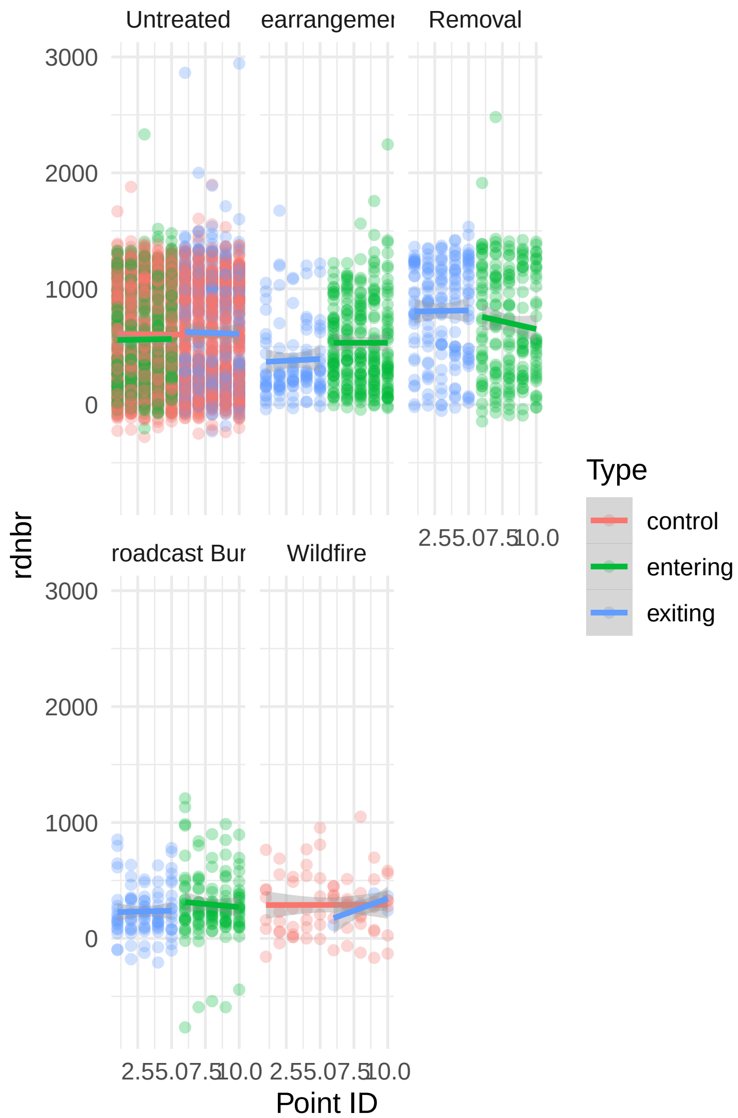
Look at shape of relationship between each predictor and RdNBR











Create linear regression model including nested random effects of transect ID and fire ID. Fixed effects include treatment characteristics and an interaction between point distance and transect type

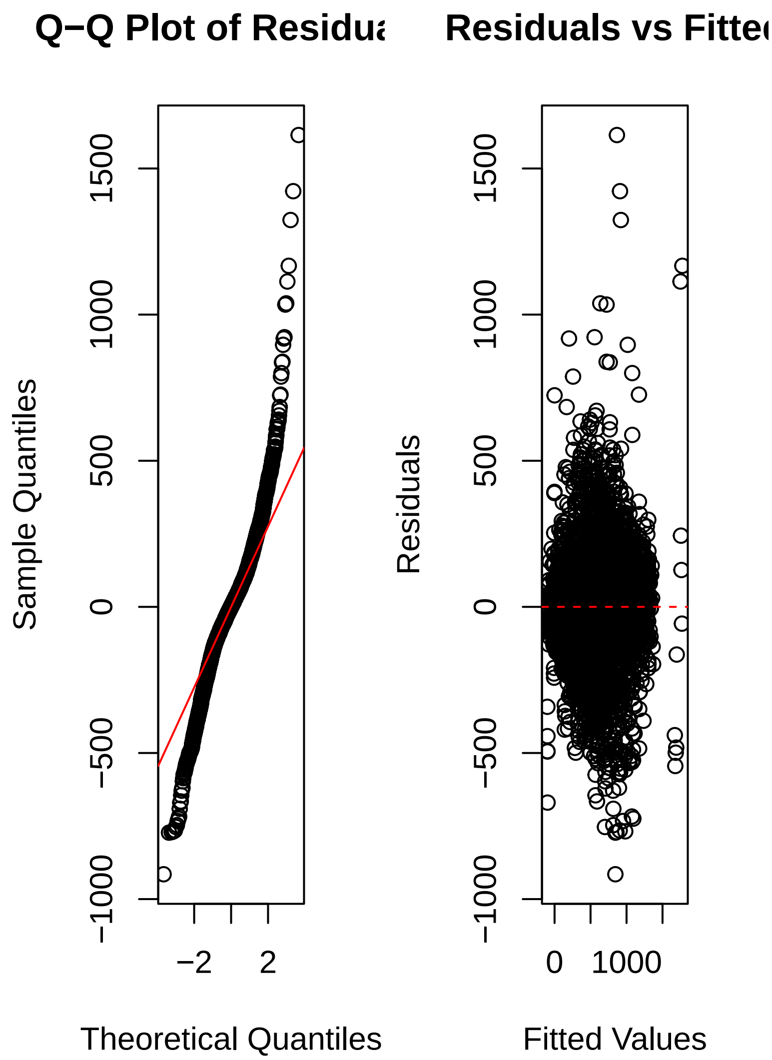
- currently not including age or prev burn sev

```
## Linear mixed model fit by REML ['lmerMod']
## Formula:
## rdnbr ~ TRT_TYPE + COUNT + Point_ID * type + (1 | Event_ID/Transect_ID)
##    Data: transects
##
## REML criterion at convergence: 53554.2
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
```

```
## -4.3997 -0.4486  0.0014  0.4455  7.7633
##
## Random effects:
##  Groups               Name        Variance Std.Dev.
##  Transect_ID:Event_ID (Intercept) 87494    295.8
##  Event_ID             (Intercept) 56733    238.2
##  Residual                         43247    208.0
## Number of obs: 3880, groups:  Transect_ID:Event_ID, 388; Event_ID, 11
##
## Fixed effects:
##                         Estimate Std. Error t value
## (Intercept)             437.7418    78.0309   5.610
## TRT_TYPERearrangement    25.7780    23.8974   1.079
## TRT_TYPERemoval           9.7843    30.9909   0.316
## TRT_TYPEBroadcast Burn  -84.7580    26.7526  -3.168
## TRT_TYPEWildfire       -205.0112    77.5522  -2.644
## COUNT                    -9.3028     4.8512  -1.918
## Point_ID                 -0.5664     1.6438  -0.345
## typeentering            -24.9895    39.1551  -0.638
## typeexiting             -63.4551    51.3955  -1.235
## Point_ID:typeentering    -0.1171     3.9639  -0.030
## Point_ID:typeexiting      7.1172     4.2493   1.675
##
## Correlation of Fixed Effects:
##             (Intr) TRT_TYPERr TRT_TYPERm TRT_TB TRT_TYPEW COUNT  Pnt_ID typntr
## TRT_TYPERrr -0.005
## TRT_TYPERmv -0.001  0.617
## TRT_TYPEBrB -0.009  0.556      0.554
## TRT_TYPEWld -0.034  0.008      0.019      0.080
## COUNT        0.008 -0.345     -0.689     -0.310 -0.040
## Point_ID    -0.116  0.000      0.000      0.000  0.000     0.000
## typeenterng -0.191  0.133      0.086      0.118  0.071     0.013  0.231
## typeexiting -0.133 -0.382     -0.321     -0.361  0.063     0.003  0.176  0.193
## Pnt_ID:typn  0.048 -0.596     -0.402     -0.509 -0.008    -0.049 -0.415 -0.376
## Pnt_ID:typx  0.047  0.520      0.438      0.495 -0.049     0.003 -0.387  0.018
##             typxtn Pnt_ID:typn
## TRT_TYPERrr
## TRT_TYPERmv
## TRT_TYPEBrB
## TRT_TYPEWld
## COUNT
## Point_ID
## typeenterng
## typeexiting
## Pnt_ID:typn  0.289
## Pnt_ID:typx -0.585 -0.337
```

```
##                    GVIF Df GVIF^(1/(2*Df))
## TRT_TYPE       8.507530  4        1.306849
## COUNT          3.648414  1        1.910082
## Point_ID       2.000000  1        1.414214
## type           2.027650  2        1.193296
## Point_ID:type 10.459270  2        1.798355
```

Evaluating model assumptions



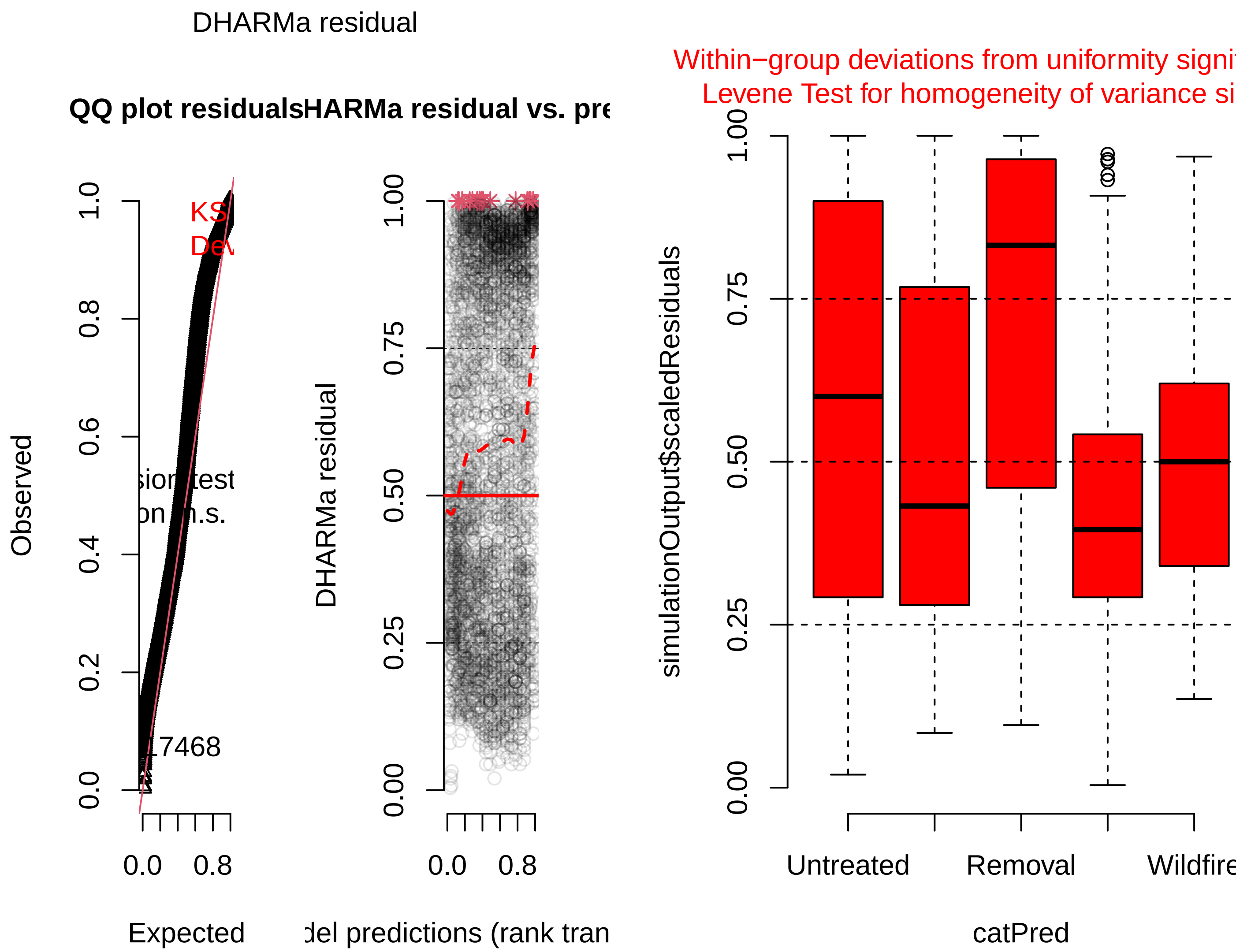
Fitted vs residuals plot

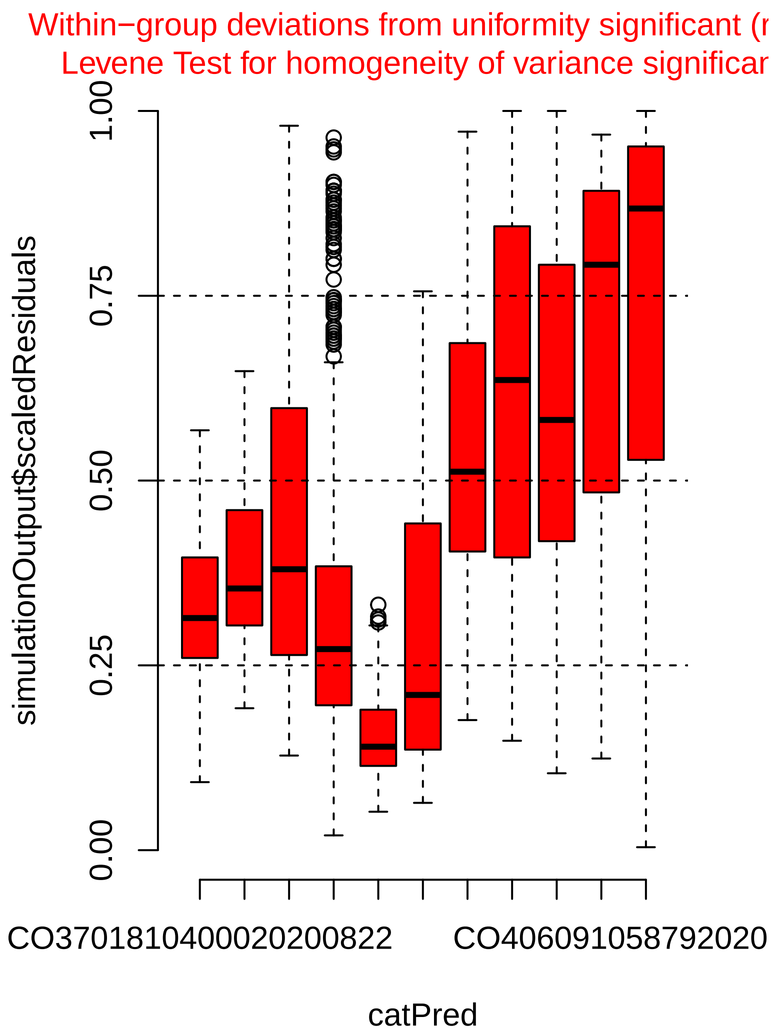
Try a model with rdnbr and weighted least squares

Or dispersion formula?

```
##  Family: gaussian  ( identity )
## Formula:
## rdnbr ~ TRT_TYPE + COUNT + Point_ID * type + (1 | Event_ID/Transect_ID)
## Dispersion:             ~Event_ID
## Data: transects
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##   53013.0   53163.3  -26482.5   52965.0      3856
##
## Random effects:
```

```
##
## Conditional model:
##  Groups                Name        Variance Std.Dev.
##  Transect_ID:Event_ID (Intercept) 86857    294.7
##  Event_ID             (Intercept) 57062    238.9
##  Residual                             NA       NA
## Number of obs: 3880, groups:  Transect_ID:Event_ID, 388; Event_ID, 11
##
## Conditional model:
##                         Estimate Std. Error z value Pr(>|z|)
## (Intercept)             430.4692    78.0649   5.514  3.5e-08 ***
## TRT_TYPERearrangement    48.6800    18.9767   2.565   0.0103 *
## TRT_TYPERemoval          34.0973    25.2926   1.348   0.1776
## TRT_TYPEBroadcast Burn  -44.9563    21.1112  -2.129   0.0332 *
## TRT_TYPEWildfire       -154.8837    73.7058  -2.101   0.0356 *
## COUNT                   -11.4576     4.4489  -2.575   0.0100 *
## Point_ID                  0.2129     1.3228   0.161   0.8721
## typeentering            -31.3785    37.8212  -0.830   0.4067
## typeexiting             -53.2815    47.8224  -1.114   0.2652
## Point_ID:typeentering    -0.6697     3.1421  -0.213   0.8312
## Point_ID:typeexiting      3.4520     3.4297   1.007   0.3142
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Dispersion model:
##                               Estimate Std. Error z value Pr(>|z|)
## (Intercept)                    4.29299    0.09745   44.05  < 2e-16 ***
## Event_IDC03717510737520170802  0.05692    0.13883    0.41    0.682
## Event_IDC03740210724320120513  0.99782    0.10687    9.34  < 2e-16 ***
## Event_IDC03765810847420180722  0.60379    0.10175    5.93 2.95e-09 ***
## Event_IDC03772010868220190620 -0.08004    0.12482   -0.64    0.521
## Event_IDC03812310818320100522  0.62047    0.12863    4.82 1.41e-06 ***
## Event_IDC03823310568320110612  0.89676    0.12843    6.98 2.90e-12 ***
## Event_IDC03905510606920180628  1.46136    0.10763   13.58  < 2e-16 ***
## Event_IDC03937110704320180703  1.21035    0.10869   11.14  < 2e-16 ***
## Event_IDC04015410538220201017  0.93898    0.10622    8.84  < 2e-16 ***
## Event_IDC04060910587920200813  1.12878    0.09922   11.38  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```


DHARMa residual
QQ plot residuals
HARMa residual vs. pre
KS
Dev
sion test
on n.s.
17468
Observed
Expected
DHARMa residual
del predictions (rank tran
Within−group deviations from uniformity signi
Levene Test for homogeneity of variance si
simulationOutput$scaledResiduals
Untreated
Removal
Wildfire
catPred



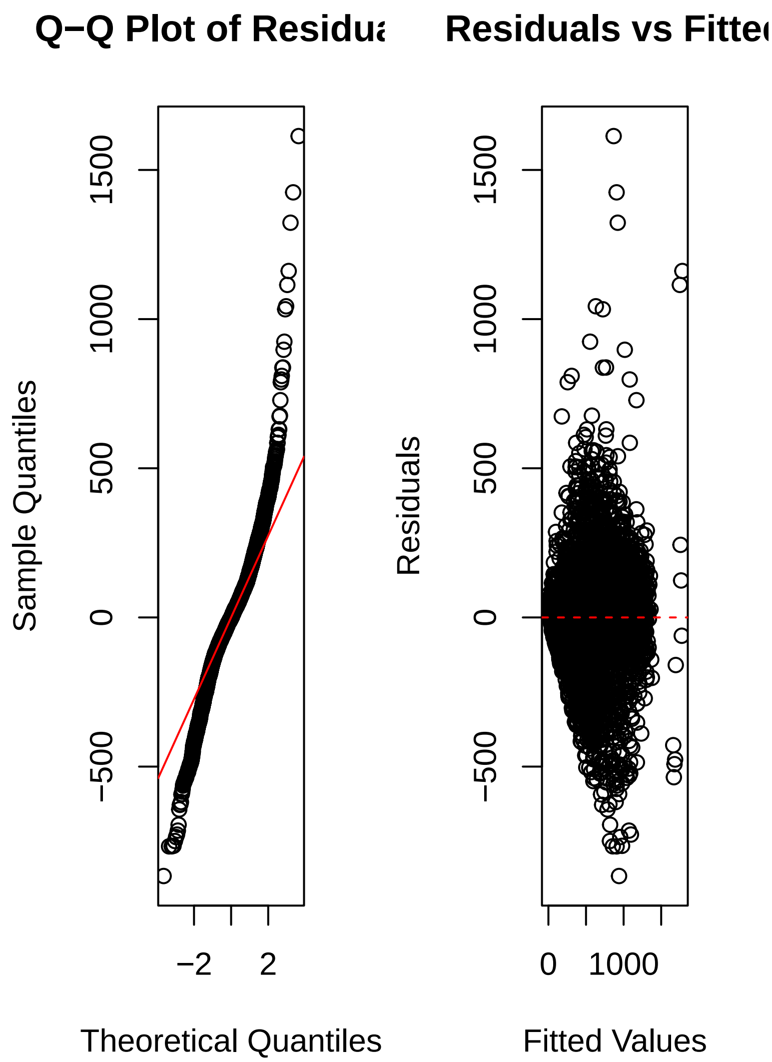
Try a model with only positive rdnbr

```
## Linear mixed model fit by REML ['lmerMod']
## Formula:
## rdnbr_pos ~ TRT_TYPE + COUNT + Point_ID * type + (1 | Event_ID/Transect_ID)
##    Data: transects
##
## REML criterion at convergence: 50449.9
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.2280 -0.4499 -0.0080  0.4522  7.8710
##
## Random effects:
##  Groups               Name        Variance Std.Dev.
##  Transect_ID:Event_ID (Intercept) 79664    282.2
```

```
##  Event_ID             (Intercept) 52005    228.0
##  Residual                         42022    205.0
## Number of obs: 3662, groups:  Transect_ID:Event_ID, 388; Event_ID, 11
##
## Fixed effects:
##                         Estimate Std. Error t value
## (Intercept)             459.5632    74.8912   6.136
## TRT_TYPERearrangement    30.4530    24.1231   1.262
## TRT_TYPERemoval          15.5498    31.1232   0.500
## TRT_TYPEBroadcast Burn  -53.0732    27.3150  -1.943
## TRT_TYPEWildfire       -206.8009    76.1208  -2.717
## COUNT                   -10.3357     4.9901  -2.071
## Point_ID                 -1.2365     1.6940  -0.730
## typeentering            -33.8178    37.9219  -0.892
## typeexiting             -84.3508    50.2800  -1.678
## Point_ID:typeentering     0.4721     4.0159   0.118
## Point_ID:typeexiting      9.5798     4.3195   2.218
##
## Correlation of Fixed Effects:
##             (Intr) TRT_TYPERr TRT_TYPERm TRT_TB TRT_TYPEW COUNT  Pnt_ID typntr
## TRT_TYPERrr -0.006
## TRT_TYPERmv -0.002  0.620
## TRT_TYPEBrB -0.010  0.555      0.554
## TRT_TYPEWld -0.033  0.010      0.021      0.085
## COUNT        0.009 -0.352     -0.685     -0.319 -0.044
## Point_ID    -0.125  0.000      0.000      0.000  0.003     0.000
## typeenterng -0.195  0.137      0.090      0.123  0.070     0.014  0.247
## typeexiting -0.135 -0.390     -0.327     -0.362  0.063    -0.005  0.186  0.186
## Pnt_ID:typn  0.053 -0.591     -0.402     -0.503 -0.010    -0.047 -0.422 -0.395
## Pnt_ID:typx  0.052  0.510      0.431      0.479 -0.053     0.011 -0.392  0.014
##             typxtn Pnt_ID:typn
## TRT_TYPERrr
## TRT_TYPERmv
## TRT_TYPEBrB
## TRT_TYPEWld
## COUNT
## Point_ID
## typeenterng
## typeexiting
## Pnt_ID:typn  0.292
## Pnt_ID:typx -0.605 -0.324

##                    GVIF Df GVIF^(1/(2*Df))
## TRT_TYPE       8.275941  4        1.302348
## COUNT          3.628428  1        1.904843
## Point_ID       2.025794  1        1.423304
## type           2.163210  2        1.212759
## Point_ID:type 10.687301  2        1.808077
```

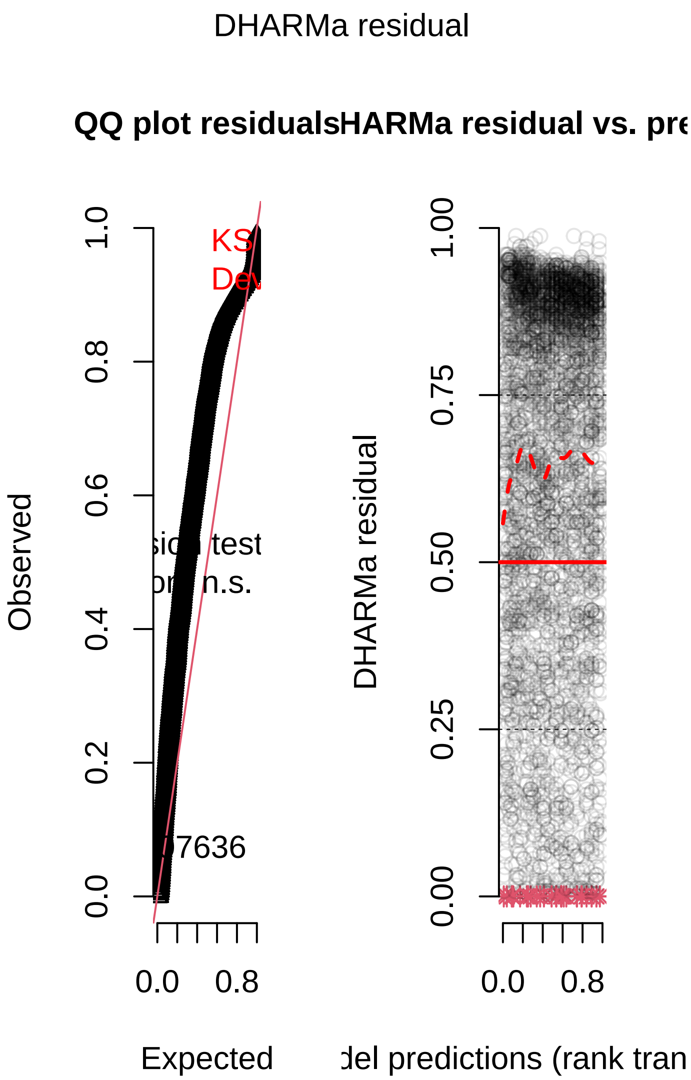
Evaluating model assumptions



Fitted vs residuals plot

Try a Gamma family

Used to model a response distribution that is positive, right skewed, and continuous


DHARMa residual
QQ plot residuals
HARMa residual vs. pre
KS
Dev
sion test
n.s.
7636
Observed
Expected
0.0
0.2
0.4
0.6
0.8
1.0
DHARMa residual
0.00
0.25
0.50
0.75
1.00
del predictions (rank tran